

Human-Computer Interation 2024/2025

Lecture Class 5

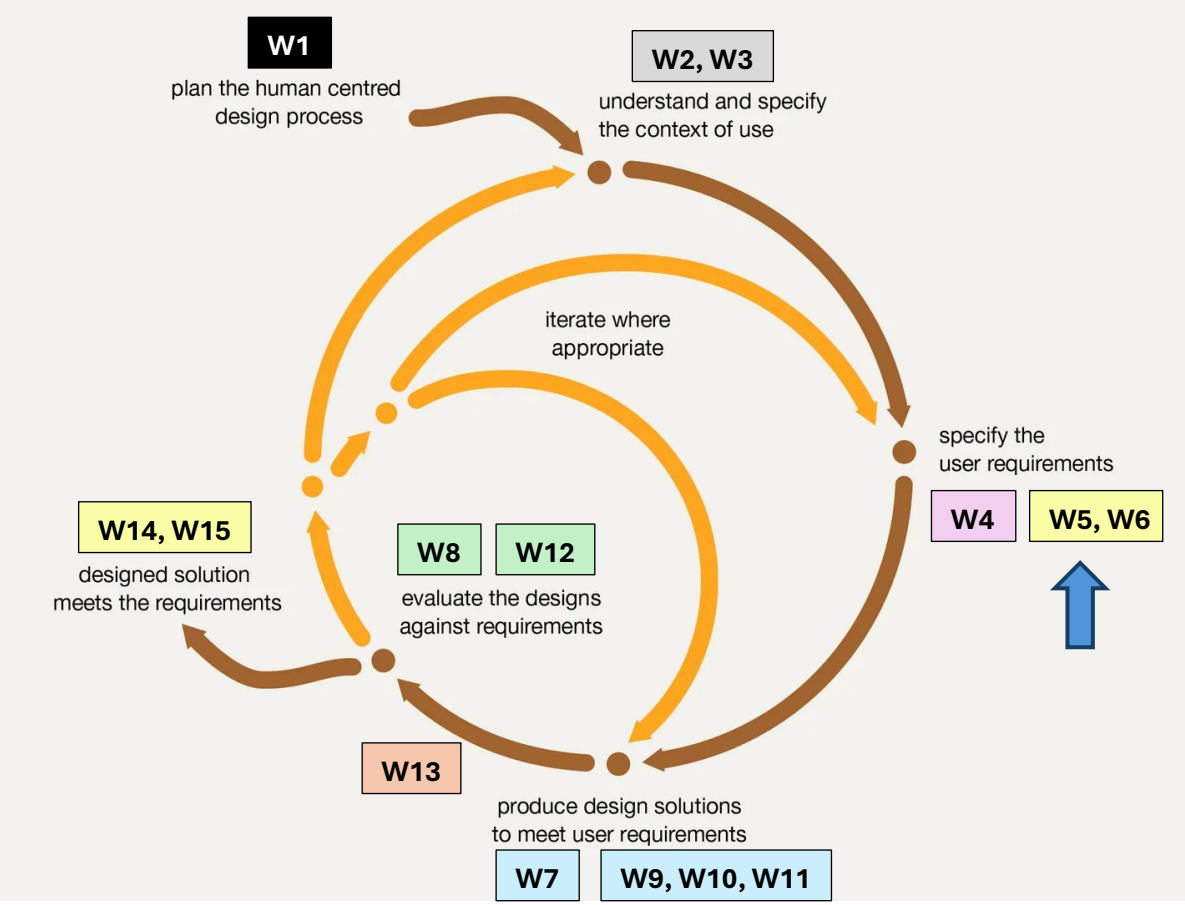
Low Fidelity Prototyping

universidade de aveiro
deti departamento de electrónica, telecomunicações e informática

This week in HCI

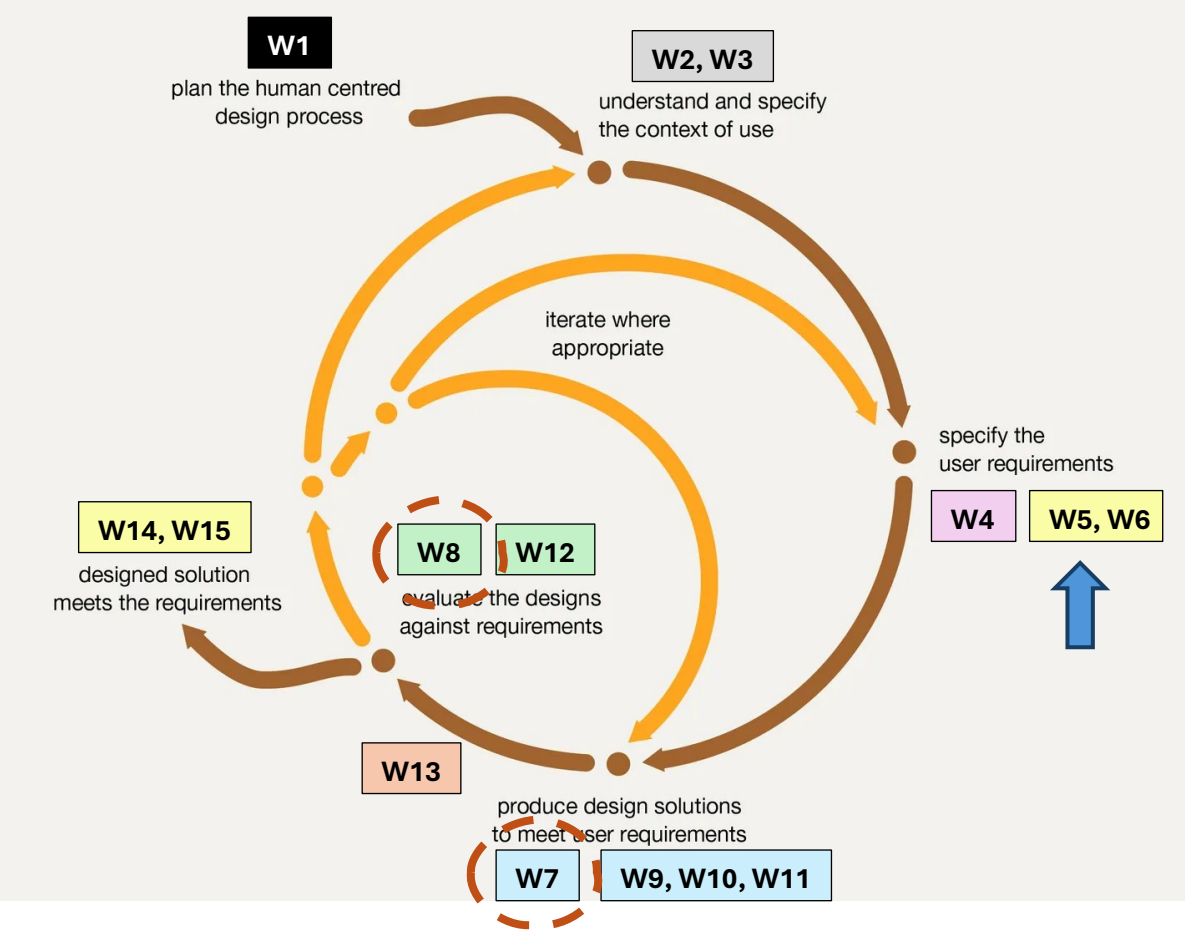
This week you present and receive feedback about the context analysis and requirements

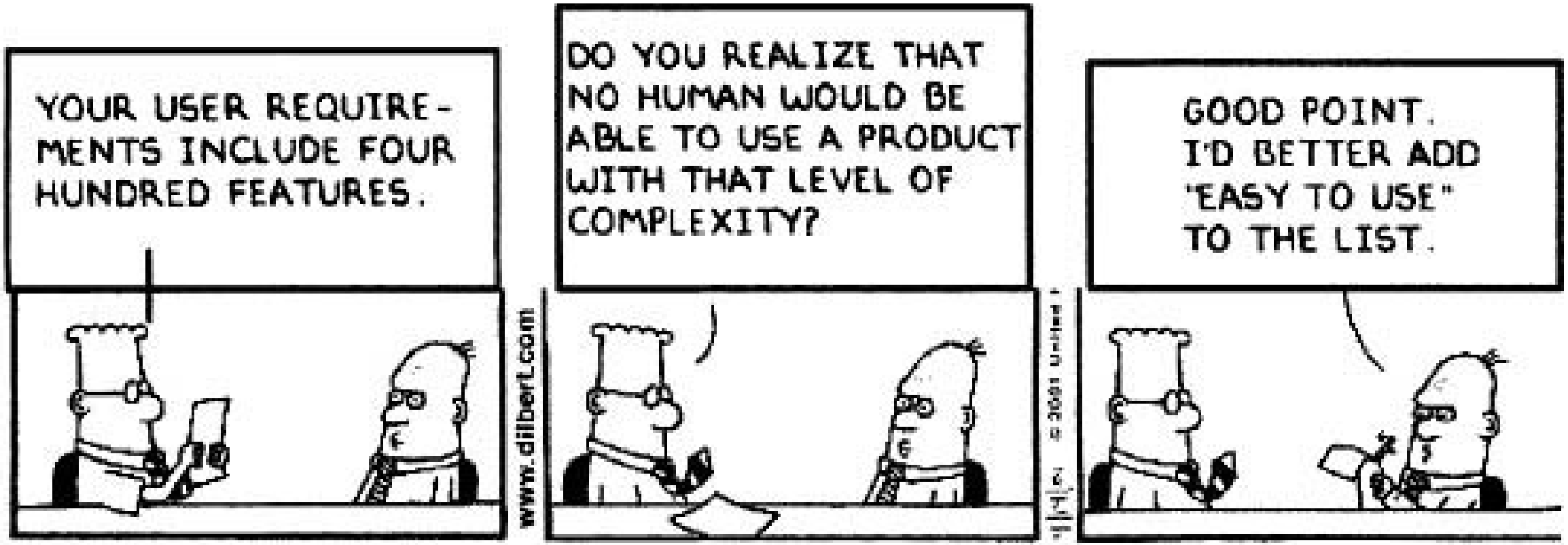
In the meantime, you will start learning to prepare the next stage…



Today's Outline

- What to do with requirements
- The usefulness of prototyping
- Article presentations





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How can we move from the requirements and start designing the interface?

We prototype!

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Thinking About Prototyping

What is a prototype?

something to test, explore, or communicate design ideas for *a thing* that is being designed.



What for?

It lets you create and refine an interface based on user feedback *before* implementing it.

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Why Should We Prototype UIs?

- Test and communicate user interface designs
- Save time and money
- Bring users into the design process
- Design across devices and platforms

What Can Prototypes Provide?

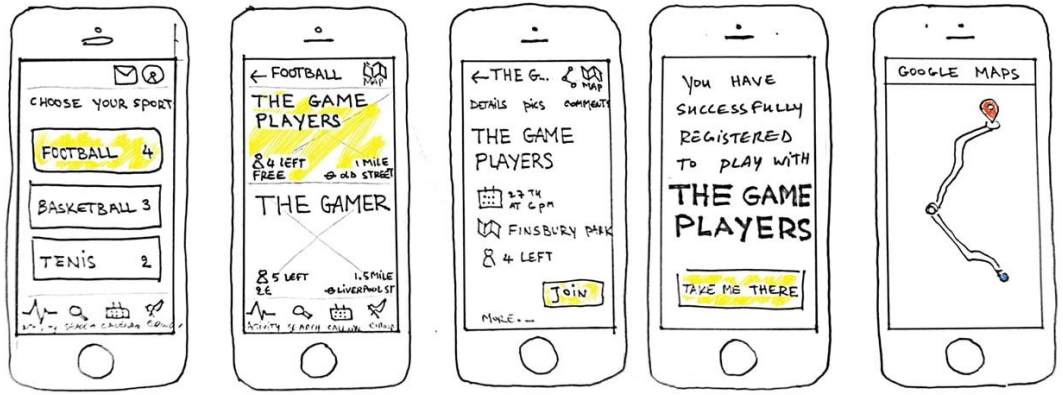
- Usability Issues
- Missing (or misspecified) functional requirements
- Preference for one design alternative
- Priorities

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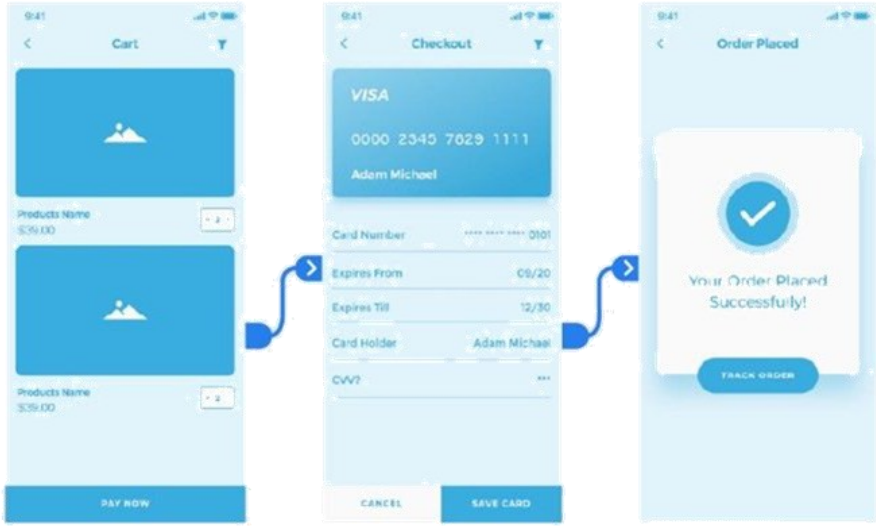
Fidelity Levels

Visual Design, Content, Interactivity

- Low Fidelity



- High Fidelity



How to Choose


https://www.lovepanky.com/wp-content/uploads/2011/09/love-or-career.jpg

- Type of prototype should match current stage of project and the questions needing an answer
- If you are just starting to think about the interface, it is probably better to go low fidelity, to have a first insight on how users react
- Go high fidelity when you already got some good feedback to back the extra investment or interaction patterns need more detail



Lo-Fi Prototyping

https://www.interaction-design.org/literature/topics/prototypes

Usefulness of Lo-Fi Prototypes

- Fast and inexpensive
- Elicit users feedback concerning general aspects
- Easy to modify even during user tests
- Valuable to test the UI conceptual model
- Estimated to allow detecting up to 80% of the usability issues

Usefulness of Lo-Fi Prototypes

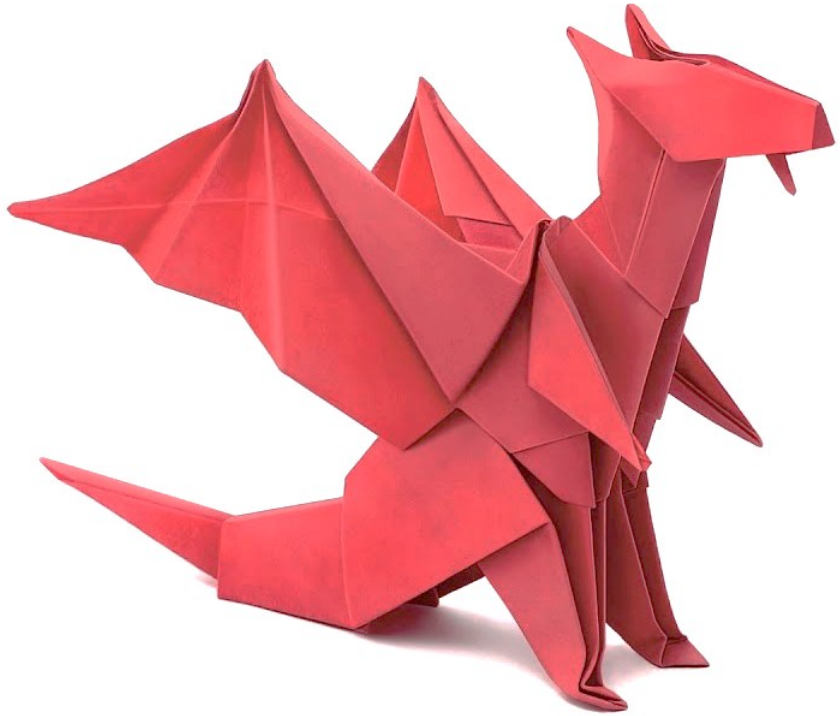
Suitable to collect data for:

Concepts and terminology
Navigation
Contents
Structure
Functionalities

Common Techniques and Tools



- Paper Prototypes
  Simple, hand-drawn representations of intefaces that allow for easy visualization and testing of navigation and functionality with minimal effort
- Wireframes
  Digital blueprints of web pages or app screens that outline the structure and layout without incorporating detailed design elements
- Digital Tools
  Software solutions enabling designers to create low-fidelity prototypes that can be easily shared and modified for collaborative feedback



Paper Prototyping

Paper prototyping is a variation of usability testing where representative users perform realistic tasks by interacting with a paper version of the interface that is manipulated by a person "playing computer," who doesn't explain how the interface is intended to work.

- Carolyn Snyder, "Paper Prototyping", ch 1

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Usefulness of Paper Prototypes

Provide a good compromise for obtaining first feedbacks and rapidly iterating over design choices



Advantages of Paper Prototypes

Quick and Cheap

Suited to Collaborating and creating a shared understanding

No training

No technical constraints



Advantages of Paper Prototypes

More Creative Feedback

Unfinished designs encourage open discussion and creativity.

Paper prototypes signal that ideas are still evolving.

Paper prototypes remove the intimidation factor of technology. They will not break anything...

Users may feel empowered to suggest changes when they see a work in progress.

Advantages of Paper Prototypes
No Nitpicky Feedback

High-fidelity prototypes invite feedback on minor visual details.

Users tend to focus on flaws when something looks "finished.“

Unwanted feedback: “Those fields don’t line up” or “I don’t like that shade of green.”

Encourages feedback on concepts and functionality rather than aesthetics.

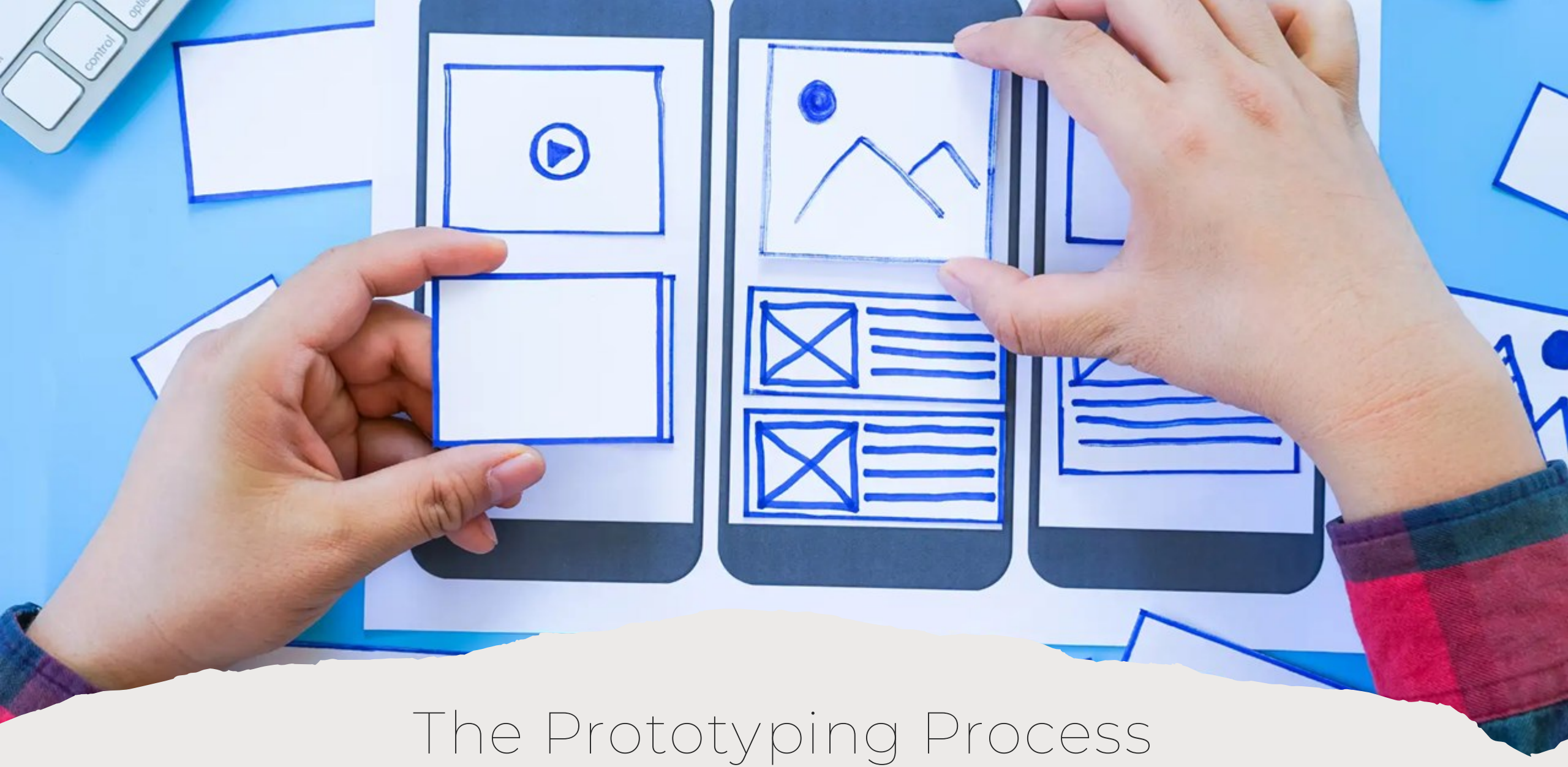
Remember...

If you show a nonprogrammer a 100% beautiful UI, they will think the system is almost done

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Disadvantages of Paper Prototypes

- Lack of Realism / unresponsive
- In-person testing with moderator
- Interpret results carefully



The Prototyping Process

Ch 5 -7

https://hiyield.co.uk/blog/building-your-first-prototype/

Task Design

- Before starting to design a prototype, think about the tasks it will support

- When asking someone to use your prototype you do not just watch them do what they want

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Guidelines to Choosing Tasks

Tasks should:

- Be based on Persona goals
- Cover questions important to the success of your system
- Have appropriate scope (not just one per screen)
- Have a finite and predictable set of possible solutions
- Have a clear end point that the user can recognize
- Elicit action, not just opinion

Choosing Tasks

Guides design decisions

- Ensures the prototype supports real user actions.

Prevents unnecessary details

- Focus on essential interactions.

Reveals missing elements

- Helps identify gaps in the workflow early.



Bad Task Examples

- Explore the website and tell us what you think.
- Use the app to complete a financial transaction.
- Pretend you are a new user and navigate through the entire platform.
- Find and read all terms and conditions on the website.
- Go to the Account Settings and change your billing address

Good Task Examples

- Find and purchase a pair of Asics running shoes under $100.
- Schedule an appointment with a dermatologist for next Tuesday at 10 AM.
- Upload the latest report in your archive and share it with Mary Poppins
- Change your default billing address to “R. da Casa, 3810 Aveiro”

Guidelines to Choosing Tasks

- Think about the order of the tasks
- Avoid wording that reveals the method
  - Bad: Go to the File Menu and Export the file to PDF
  - Good: Save the current document as a PDF
- Sometimes an image may help…



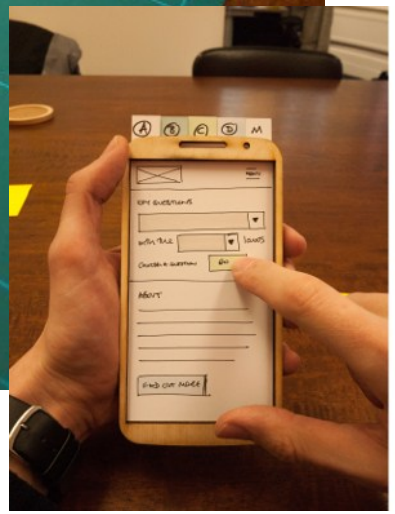


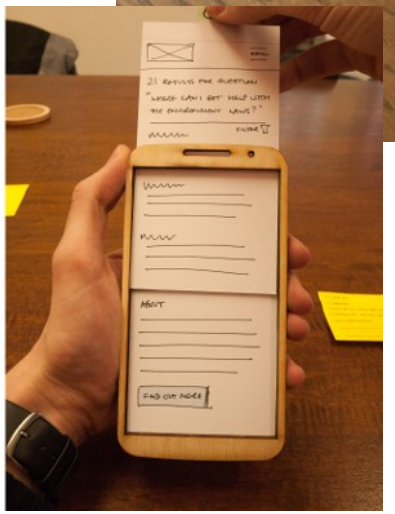
Building the Prototype

Prototype Background









Prototype Content

It does not need to have much detail, nor to be very realistic, e.g.:

Text may be replaced by some lines

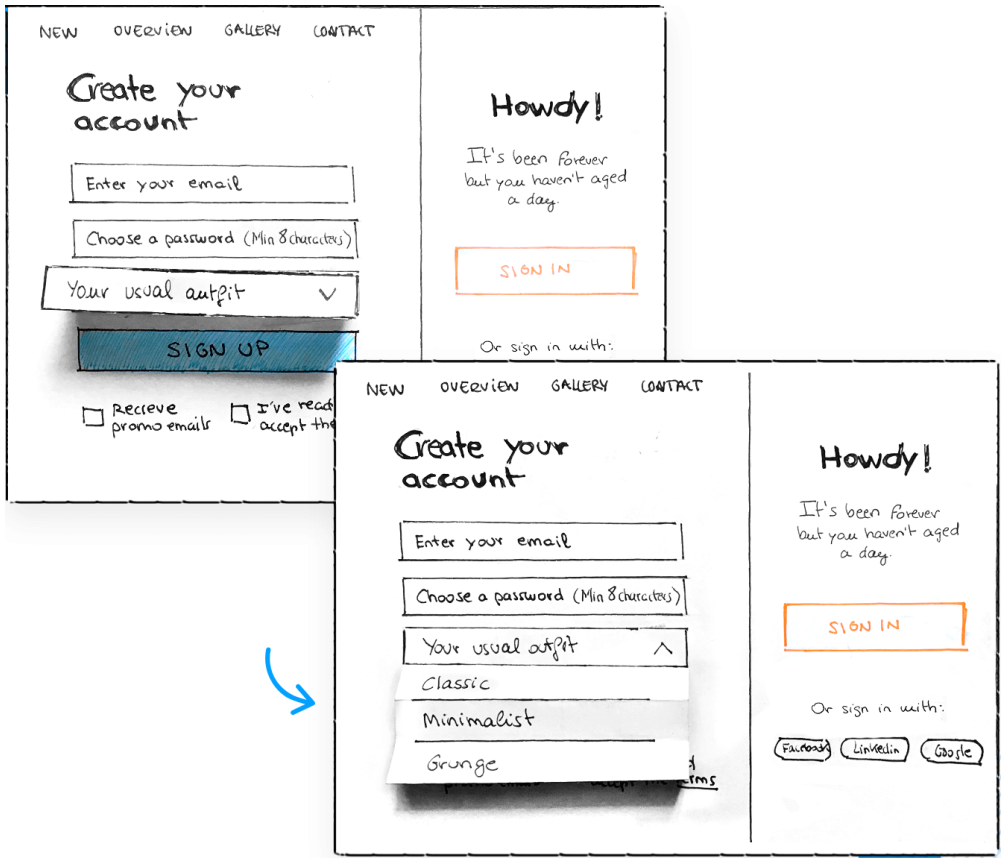
Images may be replaced by words

In general, no colour is needed

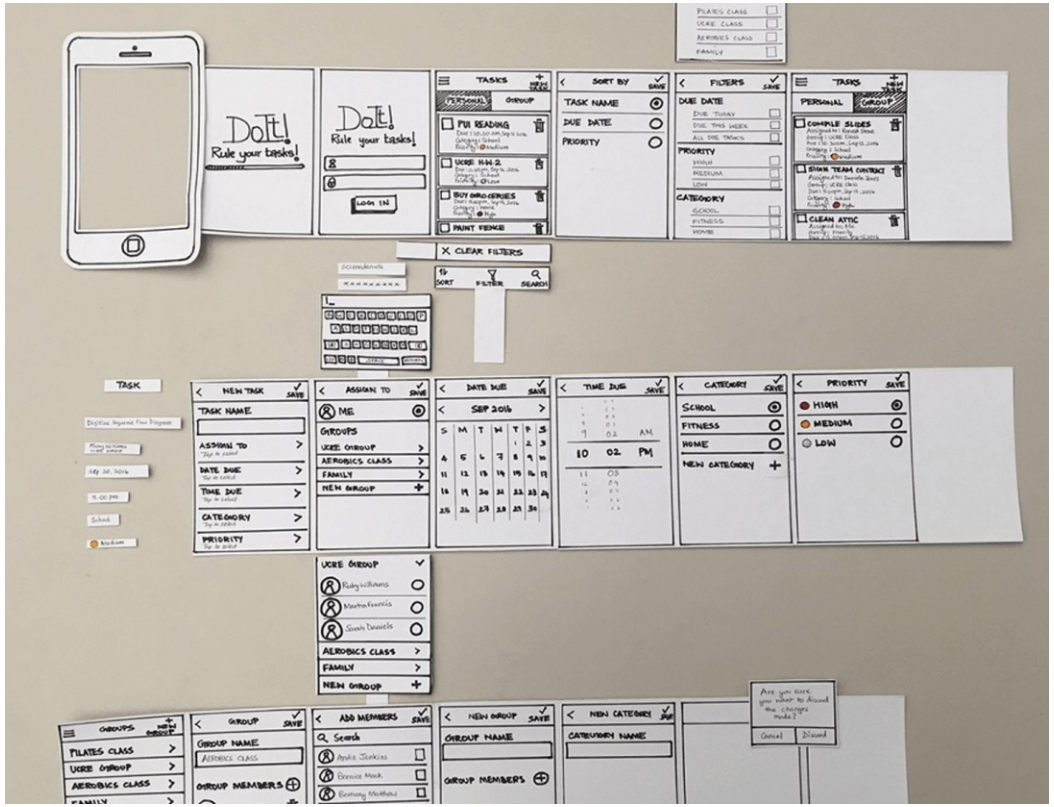
Sizes of windows, fonts, etc. don't need to be final



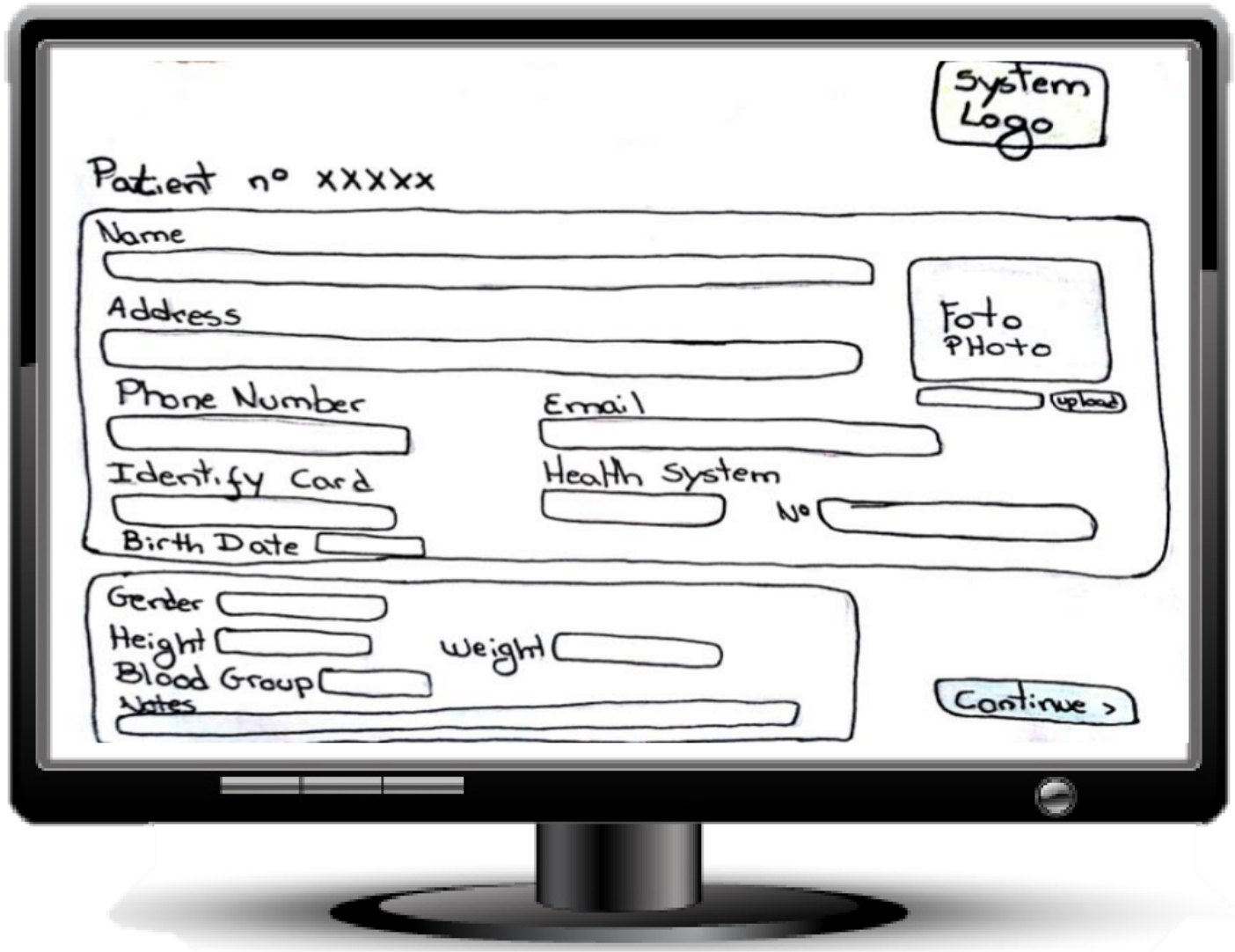
Examples


NEW
OVERVIEW
GALLERY
CONTACT
Create your account
Enter your email
Choose a password (Min 8 characters)
Your usual outfit
SIGN UP
Recieve promo emails
Howdy!
It's been forever but you haven't aged a day.
SIGN IN
Or sign in with:
Classic
Minimalist
Grunge


Sign up
Name
E-mail
Password
Confirm Password
Sign up
You already have login?


DoIt!
Rule your tasks!
LOG IN
TASKS
SORT BY
FILTERS
CLEAR FILTERS
NEW TASK
ASSIGN TO
DATE DUE
TIME DUE
CATEGORY
PRIORITY
GROUPS
NEW GROUP
ADD MEMBERS
NEW CATEGORY

e-DOCTOR


System Logo
Patient nº xxxxx
Name
Address
Foto
PHoto
upload
Phone Number
Email
Identify Card
Health System
Nº
Birth Date
Gender
Height
Weight
Blood Group
Notes
Continue >



Evaluating a Prototype

https://blog.adobe.com/en/publish/2017/11/29/prototyping-difference-low-fidelity-high-fidelity-prototypes-use#what-is-fidelity

Remember the Tasks?

The tasks considered for designing the prototype are very useful for testing:

- Improves usability testing
  Users interact meaningfully instead of guessing functionality.
- Encourages realistic feedback
  Users evaluate based on practical use cases, not abstract ideas.

Using Tasks during Testing

- Be attentive to task order
  - Think about a logical order. Sometimes tasks provide a narrative/flow that can help users understand the logic
- Present tasks in written form, not orally
  - This allows users to get back to the task and analyse it further or remember some detail
- Present one task at a time
  - If you present several tasks, it is possible that users start reading next tasks before doing the current one, and this may influence their actions

Who is at the Test Session?

- The computer
- The facilitator
- The observer

… and the user



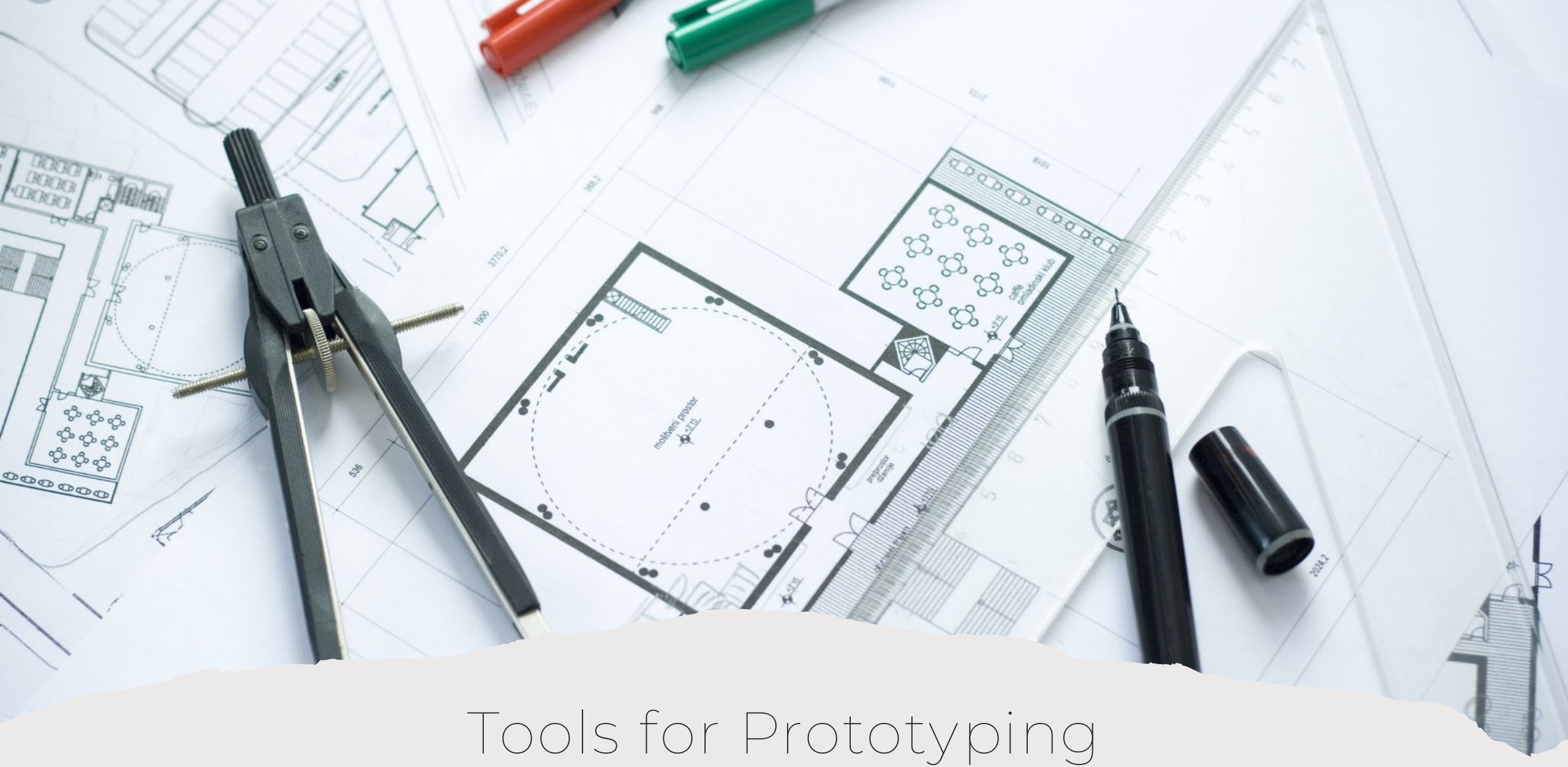
Anatomy of a Test Session

- Receive the participant and explain the overall context of your system
- Give the participant the different tasks, one by one
- Observe the participant using the prototype and collect their feedback
- Encourage thinking aloud
- Discuss opinions and ideas at the end

Prototype Evaluation Example



- https://www.youtube.com/watch?v=OIbdIXLunt4&t=7s&ab_channel=NNgroup



Tools for Prototyping

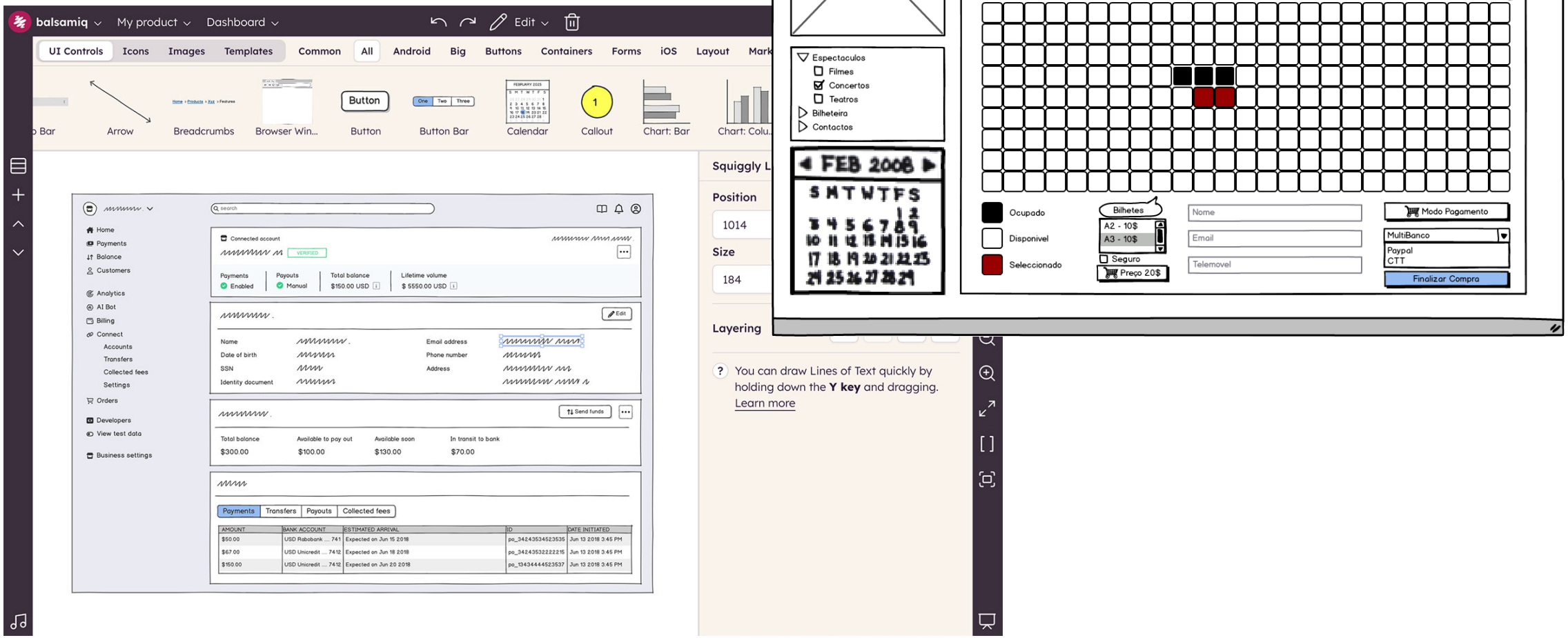
Beyond sketched paper prototypes

Prototyping beyond paper

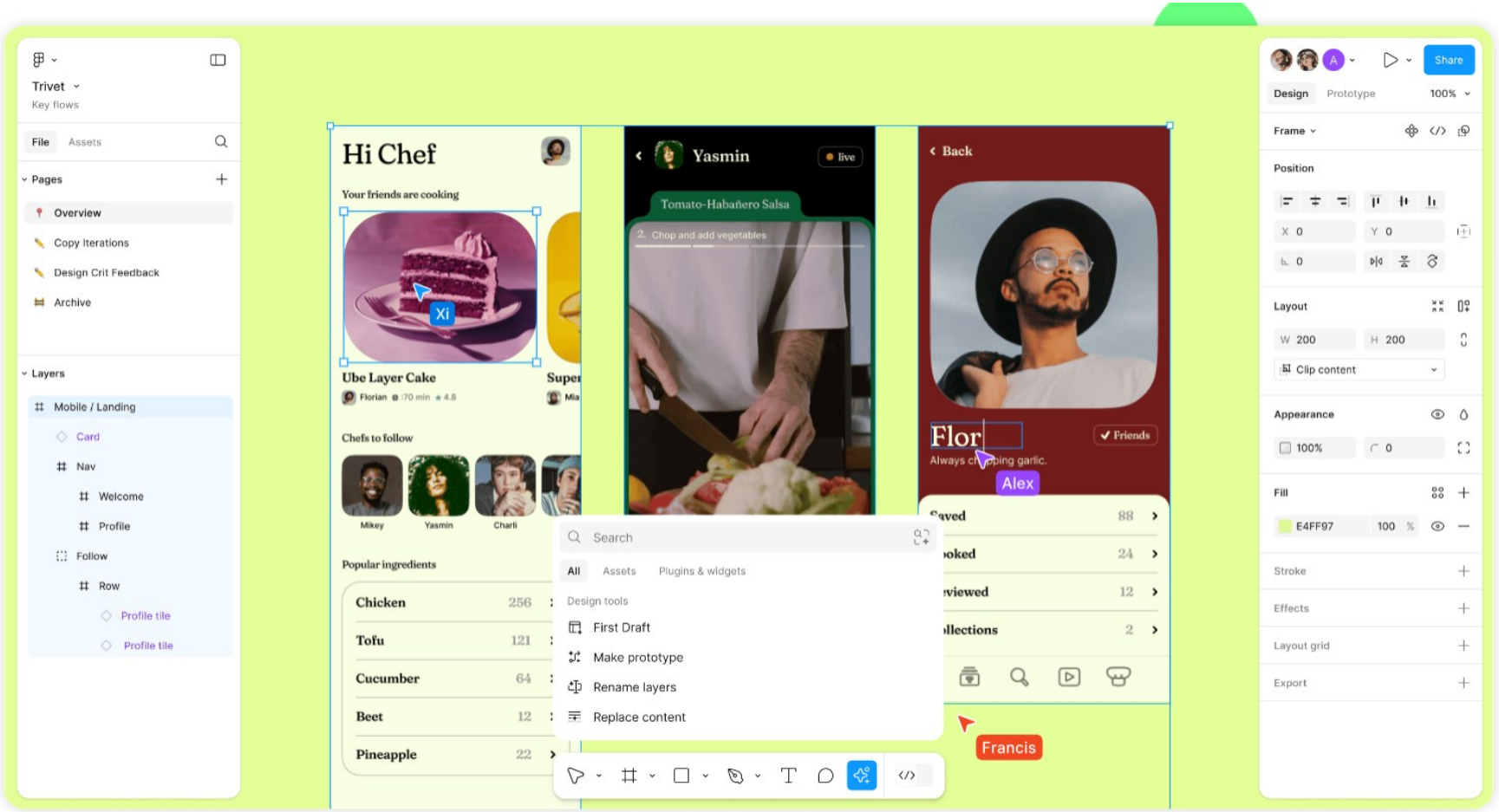
- The procedures mentioned for paper prototypes can be performed beyond rough sketches
- Sometimes, the use of tools beyond pen and paper can yield increased interactivity and improved look and feel

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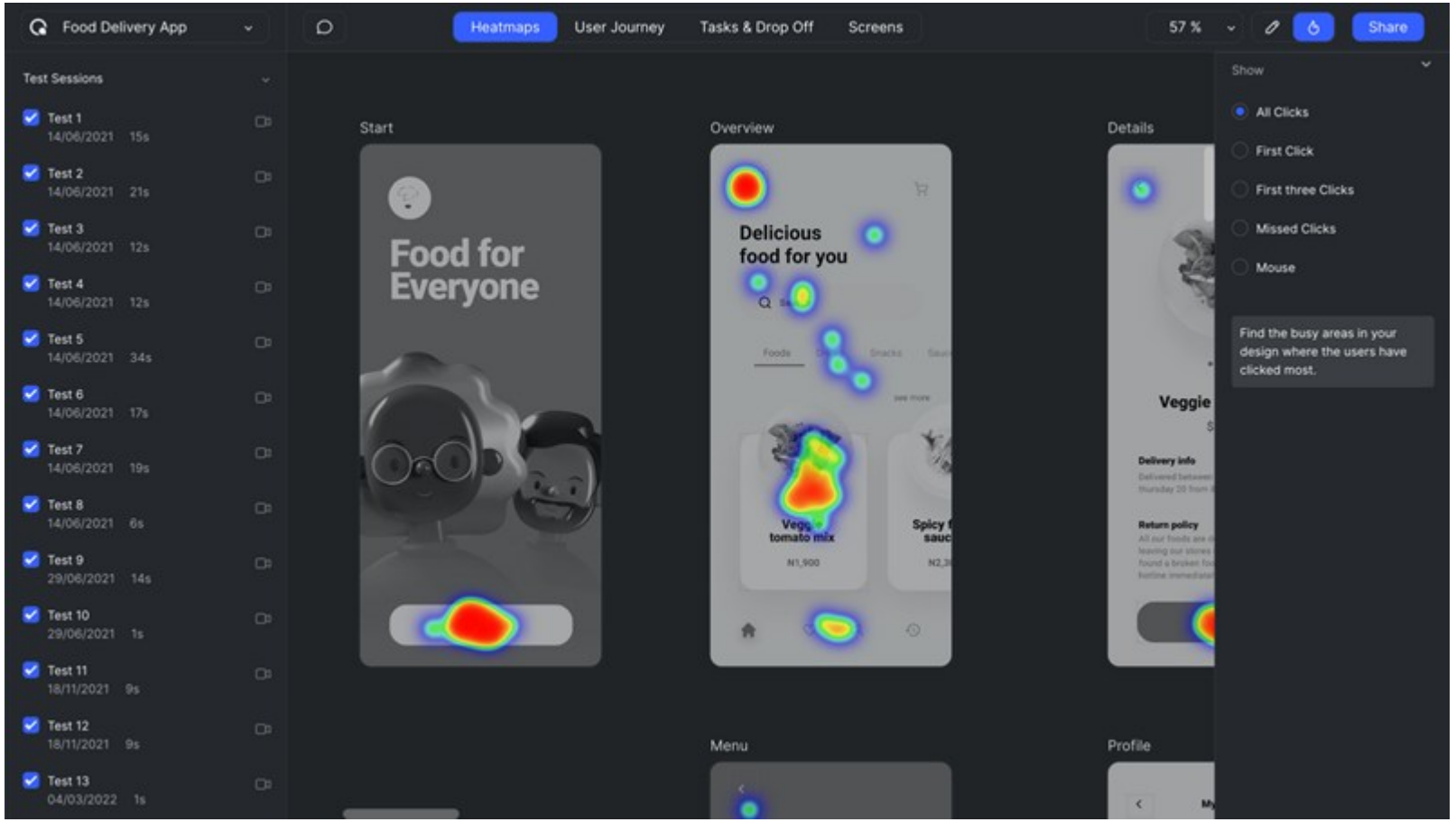
Balsamiq


balsamiq
My product
Dashboard
Edit
UI Controls
Icons
Images
Templates
Common
All
Android
Big
Buttons
Containers
Forms
iOS
Layout
Arrow
Breadcrumbs
Browser Win...
Button
Button Bar
Calendar
Callout
Chart: Bar
Squiggly L
Position
1014
Size
184
Layering
You can draw Lines of Text quickly by holding down the Y key and dragging.
Learn more
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http://cine-teatro-estarreja.pt/ticket
LOGO DO CINE-TEATRO
Imagens dos espectaculos a passar
search
Bilheteira
Espectaculos
Filmes
Concertos
Teatros
Bilheteira
Contactos
FEB 2008
Ocupado
Disponivel
Seleccionado
Bilhetes
A2 - 10$
A3 - 10$
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Nome
Email
Telemovel
Modo Pagamento
MultiBanco
Paypal
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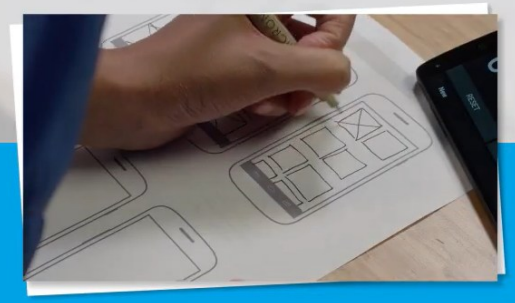
Figma


Trivet
Key flows
File
Assets
Pages
Overview
Copy Iterations
Design Crit Feedback
Archive
Layers
Mobile / Landing
Card
Nav
Welcome
Profile
Follow
Row
Profile tile
Profile tile
Hi Chef
Your friends are cooking
Ube Layer Cake
Chefs to follow
Mikey
Yasmin
Charli
Popular ingredients
Chicken 256
Tofu 121
Cucumber 64
Beet 12
Pineapple 22
Yasmin
live
Tomato-Habañero Salsa
2. Chop and add vegetables
Back
Flor
Friends
Always chopping garlic.
Alex
Saved 88
Liked 24
Viewed 12
Collections 2
Search
All
Assets
Plugins & widgets
Design tools
First Draft
Make prototype
Rename layers
Replace content
Francis
Share
Design
Prototype
100%
Frame
Position
X 0
Y 0
Layout
W 200
H 200
Clip content
Appearance
100%
Fill
E4FF97
100 %
Stroke
Effects
Layout grid
Export

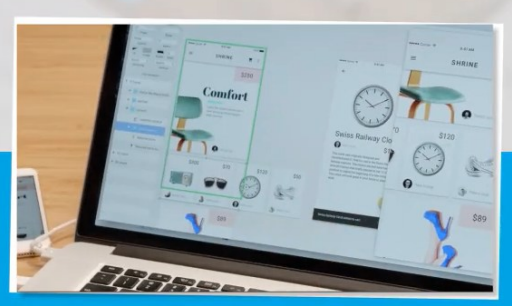
Quant UX


Food Delivery App
Heatmaps
User Journey
Tasks & Drop Off
Screens
57 %
Share
Test Sessions
Test 1 14/06/2021 15s
Test 2 14/06/2021 21s
Test 3 14/06/2021 12s
Test 4 14/06/2021 12s
Test 5 14/06/2021 34s
Test 6 14/06/2021 17s
Test 7 14/06/2021 19s
Test 8 14/06/2021 6s
Test 9 29/06/2021 14s
Test 10 29/06/2021 1s
Test 11 18/11/2021 9s
Test 12 18/11/2021 9s
Test 13 04/03/2022 1s
Start
Food for Everyone
Overview
Delicious food for you
Details
Veggie
Menu
Profile
Show
All Clicks
First Click
First three Clicks
Missed Clicks
Mouse
Find the busy areas in your design where the users have clicked most.

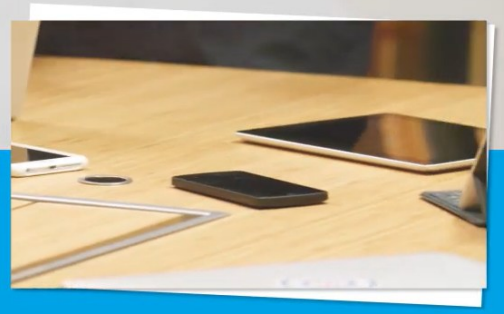
Rapid Prototyping



Sketching & Paper Prototyping



Digital Prototyping



Native Prototyping

Rapid Prototyping by Google for Startups:
https://www.youtube.com/watch?v=JMjozqJS44M
https://www.youtube.com/watch?v=KWGBGTGryFk
https://www.youtube.com/watch?v=IusOgox4xMI

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